MSG No. 129.420 P_c4/nmm [Type IV, tetragonal]

Table 1: Wyckoff site: `4a`, site symmetry: `-4m2`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[1,6,7,8,10,11,12,13]`
2	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[2,3,4,5,9,14,15,16]`
3	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[17,22,23,24,26,27,28,29]`
4	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[18,19,20,21,25,30,31,32]`

Table 2: Wyckoff site: `4b`, site symmetry: `-4'm2'`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[1,6,12,13,23,24,26,27]`
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[2,3,15,16,20,21,25,30]`
3	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[4,5,9,14,18,19,31,32]`
4	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[7,8,10,11,17,22,28,29]`

Table 3: Wyckoff site: `4c`, site symmetry: `4mm`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	`[1,2,3,6,12,13,15,16]`
2	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	`[4,5,7,8,9,10,11,14]`
3	$\left[\frac{1}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	`[17,18,19,22,28,29,31,32]`
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	`[20,21,23,24,25,26,27,30]`

Table 4: Wyckoff site: `8d`, site symmetry: `..2/m`

No.	position	mapping
1	$[0, 0, 0]$	`[1,8,9,16]`
2	$\left[\frac{1}{2}, 0, 0\right]$	`[2,4,10,12]`
3	$\left[0, \frac{1}{2}, 0\right]$	`[3,5,11,13]`
4	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[6,7,14,15]`
5	$\left[0, 0, \frac{1}{2}\right]$	`[17,24,25,32]`
6	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[18,20,26,28]`
7	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[19,21,27,29]`
8	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[22,23,30,31]`

Table 5: Wyckoff site: `8e`, site symmetry: `..2'/m`

No.	position	mapping
1	$\left[0, 0, \frac{1}{4}\right]$	`[1,16,24,25]`
2	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[2,12,20,26]`
3	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[3,13,21,27]`
4	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[4,10,18,28]`

continued ...

Table 5

No.	position	mapping
5	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	[5,11,19,29]
6	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	[6,15,23,30]
7	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	[7,14,22,31]
8	$\left[0,\ 0,\ \frac{3}{4}\right]$	[8,9,17,32]

Table 6: Wyckoff site: 8f, site symmetry: 2mm.

No.	position	mapping
1	$\left[\frac{3}{4},\ \frac{1}{4},\ z\right]$	[1,6,12,13]
2	$\left[\frac{1}{4},\ \frac{3}{4},\ z\right]$	[2,3,15,16]
3	$\left[\frac{1}{4},\ \frac{3}{4},\ -z\right]$	[4,5,9,14]
4	$\left[\frac{3}{4},\ \frac{1}{4},\ -z\right]$	[7,8,10,11]
5	$\left[\frac{3}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	[17,22,28,29]
6	$\left[\frac{1}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	[18,19,31,32]
7	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	[20,21,25,30]
8	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	[23,24,26,27]

Table 7: Wyckoff site: 16g, site symmetry: ..2

No.	position	mapping
1	$[x,\ -x,\ 0]$	[1,8]
2	$\left[x+\frac{1}{2},\ x,\ 0\right]$	[2,4]
3	$\left[-x,\ \frac{1}{2}-x,\ 0\right]$	[3,5]
4	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ 0\right]$	[6,7]
5	$[-x,\ x,\ 0]$	[9,16]
6	$\left[\frac{1}{2}-x,\ -x,\ 0\right]$	[10,12]
7	$\left[x,\ x+\frac{1}{2},\ 0\right]$	[11,13]
8	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$	[14,15]
9	$\left[x,\ -x,\ \frac{1}{2}\right]$	[17,24]
10	$\left[x+\frac{1}{2},\ x,\ \frac{1}{2}\right]$	[18,20]
11	$\left[-x,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[19,21]
12	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[22,23]
13	$\left[-x,\ x,\ \frac{1}{2}\right]$	[25,32]
14	$\left[\frac{1}{2}-x,\ -x,\ \frac{1}{2}\right]$	[26,28]
15	$\left[x,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[27,29]
16	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[30,31]

Table 8: Wyckoff site: 16h, site symmetry: ..2'

No.	position	mapping
1	$\left[x,\ -x,\ \frac{1}{4}\right]$	[1,24]
2	$\left[x+\frac{1}{2},\ x,\ \frac{1}{4}\right]$	[2,20]
3	$\left[-x,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[3,21]

continued ...

Table 8

No.	position	mapping
4	$\left[x+\frac{1}{2},\, x,\, \frac{3}{4}\right]$	[4,18]
5	$\left[-x,\, \frac{1}{2}-x,\, \frac{3}{4}\right]$	[5,19]
6	$\left[\frac{1}{2}-x,\, x+\frac{1}{2},\, \frac{1}{4}\right]$	[6,23]
7	$\left[\frac{1}{2}-x,\, x+\frac{1}{2},\, \frac{3}{4}\right]$	[7,22]
8	$\left[x,\, -x,\, \frac{3}{4}\right]$	[8,17]
9	$\left[-x,\, x,\, \frac{3}{4}\right]$	[9,32]
10	$\left[\frac{1}{2}-x,\, -x,\, \frac{3}{4}\right]$	[10,28]
11	$\left[x,\, x+\frac{1}{2},\, \frac{3}{4}\right]$	[11,29]
12	$\left[\frac{1}{2}-x,\, -x,\, \frac{1}{4}\right]$	[12,26]
13	$\left[x,\, x+\frac{1}{2},\, \frac{1}{4}\right]$	[13,27]
14	$\left[x+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{3}{4}\right]$	[14,31]
15	$\left[x+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{4}\right]$	[15,30]
16	$\left[-x,\, x,\, \frac{1}{4}\right]$	[16,25]

Table 9: Wyckoff site: 16i, site symmetry: .m.

No.	position	mapping
1	$\left[\frac{1}{4},\, y,\, z\right]$	[1,12]
2	$\left[\frac{1}{2}-y,\, \frac{1}{4},\, z\right]$	[2,15]
3	$\left[y,\, \frac{1}{4},\, z\right]$	[3,16]
4	$\left[\frac{3}{4},\, -y,\, -z\right]$	[4,9]
5	$\left[\frac{3}{4},\, y+\frac{1}{2},\, -z\right]$	[5,14]
6	$\left[\frac{1}{4},\, \frac{1}{2}-y,\, z\right]$	[6,13]
7	$\left[y+\frac{1}{2},\, \frac{3}{4},\, -z\right]$	[7,10]
8	$\left[-y,\, \frac{3}{4},\, -z\right]$	[8,11]
9	$\left[\frac{1}{4},\, y,\, z+\frac{1}{2}\right]$	[17,28]
10	$\left[\frac{1}{2}-y,\, \frac{1}{4},\, z+\frac{1}{2}\right]$	[18,31]
11	$\left[y,\, \frac{1}{4},\, z+\frac{1}{2}\right]$	[19,32]
12	$\left[\frac{3}{4},\, -y,\, \frac{1}{2}-z\right]$	[20,25]
13	$\left[\frac{3}{4},\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[21,30]
14	$\left[\frac{1}{4},\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[22,29]
15	$\left[y+\frac{1}{2},\, \frac{3}{4},\, \frac{1}{2}-z\right]$	[23,26]
16	$\left[-y,\, \frac{3}{4},\, \frac{1}{2}-z\right]$	[24,27]

Table 10: Wyckoff site: 16j, site symmetry: ..m

No.	position	mapping
1	$[x,\, x,\, z]$	[1,16]
2	$\left[\frac{1}{2}-x,\, x,\, z\right]$	[2,12]
3	$\left[x,\, \frac{1}{2}-x,\, z\right]$	[3,13]
4	$\left[x+\frac{1}{2},\, -x,\, -z\right]$	[4,10]
5	$\left[-x,\, x+\frac{1}{2},\, -z\right]$	[5,11]
6	$\left[\frac{1}{2}-x,\, \frac{1}{2}-x,\, z\right]$	[6,15]
7	$\left[x+\frac{1}{2},\, x+\frac{1}{2},\, -z\right]$	[7,14]

continued ...

Table 10

No.	position	mapping
8	$[-x,\ -x,\ -z]$	[8,9]
9	$\left[x,\ x,\ z+\frac{1}{2}\right]$	[17,32]
10	$\left[\frac{1}{2}-x,\ x,\ z+\frac{1}{2}\right]$	[18,28]
11	$\left[x,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[19,29]
12	$\left[x+\frac{1}{2},\ -x,\ \frac{1}{2}-z\right]$	[20,26]
13	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[21,27]
14	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[22,31]
15	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[23,30]
16	$\left[-x,\ -x,\ \frac{1}{2}-z\right]$	[24,25]

Table 11: Wyckoff site: 32k, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$\left[\frac{1}{2}-y,\ x,\ z\right]$	[2]
3	$\left[y,\ \frac{1}{2}-x,\ z\right]$	[3]
4	$\left[x+\frac{1}{2},\ -y,\ -z\right]$	[4]
5	$\left[-x,\ y+\frac{1}{2},\ -z\right]$	[5]
6	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	[6]
7	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ -z\right]$	[7]
8	$[-y,\ -x,\ -z]$	[8]
9	$[-x,\ -y,\ -z]$	[9]
10	$\left[y+\frac{1}{2},\ -x,\ -z\right]$	[10]
11	$\left[-y,\ x+\frac{1}{2},\ -z\right]$	[11]
12	$\left[\frac{1}{2}-x,\ y,\ z\right]$	[12]
13	$\left[x,\ \frac{1}{2}-y,\ z\right]$	[13]
14	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ -z\right]$	[14]
15	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ z\right]$	[15]
16	$[y,\ x,\ z]$	[16]
17	$\left[x,\ y,\ z+\frac{1}{2}\right]$	[17]
18	$\left[\frac{1}{2}-y,\ x,\ z+\frac{1}{2}\right]$	[18]
19	$\left[y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[19]
20	$\left[x+\frac{1}{2},\ -y,\ \frac{1}{2}-z\right]$	[20]
21	$\left[-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[21]
22	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[22]
23	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[23]
24	$\left[-y,\ -x,\ \frac{1}{2}-z\right]$	[24]
25	$\left[-x,\ -y,\ \frac{1}{2}-z\right]$	[25]
26	$\left[y+\frac{1}{2},\ -x,\ \frac{1}{2}-z\right]$	[26]
27	$\left[-y,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[27]
28	$\left[\frac{1}{2}-x,\ y,\ z+\frac{1}{2}\right]$	[28]
29	$\left[x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[29]
30	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[30]
31	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[31]
32	$\left[y,\ x,\ z+\frac{1}{2}\right]$	[32]